

Fleet Reliability Report

Contract Number CN-2024-118
Customer All Customers
Train Type All Train Types
Period Covered All Periods
Project Manager L. Moretti
Issued By Nazanin Homayoun,
Reliability Data Analyst
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Availability-Based Contract

Under this contract, Alstom is committed to guaranteeing a defined level of fleet availability for the covered vehicles, with maintenance activities planned and executed to meet the agreed availability and reliability targets over the contract period.

42	50%	0%	21
Total Tickets Closed	Confirmed Permanent Failures	Component Substitutions	Trains Covered

1. Executive Summary

Key figures for the selected customer, train type and reporting period, summarizing overall fleet reliability performance.

Tickets by Failure Cause

Causa del Guasto	Tickets	% of Total
Reproducibile – Permanente	21	50.0%
NFF	11	26.2%
Reproducibile – Non Permanente	10	23.8%

Analysis: "Reproducibile – Permanente" is the leading classification, covering 50.0% of closed tickets and marking where diagnostic effort is currently concentrated. Confirmed permanent failures make up 50.0% of the total — the portion that required a durable repair rather than a reset or monitoring — while 0.0% could not be reproduced during diagnostics, a share worth watching if it grows month over month. Together these ratios give an early signal of whether the fleet's failure mix is shifting toward more serious, hands-on interventions.

Tickets by Effect on Service

Effect on Service	Tickets	% of Total
Service stop	14	33.3%
Delay > 15 min	10	23.8%
No effect on service	10	23.8%
Speed restriction	8	19.0%

Analysis: "Service stop" is the most frequently recorded effect on service, accounting for 33.3% of all tickets in the selected period. The remaining 3 effect categories spread across less service-critical outcomes, which is the healthier pattern for fleet availability. A rising share of service-stopping effects over successive reports would be the clearest early warning of a fleet-wide reliability issue.

2. Failure Cause Breakdown

Distribution of tickets across the four failure-cause categories, classified from problem description and corrective action text.



Analysis: "Reproducible – Permanente" is the leading classification, covering 50.0% of closed tickets and marking where diagnostic effort is currently concentrated. Confirmed permanent failures make up 50.0% of the total — the portion that required a durable repair rather than a reset or monitoring — while 0.0% could not be reproduced during diagnostics, a share worth watching if it grows month over month. Together these ratios give an early signal of whether the fleet's failure mix is shifting toward more serious, hands-on interventions.

Breakdown by Train Type

Train Type	NFF	Non-Perm.	Permanent	Total
CS-200	4	6	11	21
RG-450	7	4	10	21

Analysis: Train type CS-200 shows the highest share of confirmed permanent failures relative to its own ticket volume, at 52.4% of its 21 tickets. Comparing this rate across train types — rather than raw ticket counts alone — highlights which fleet segment is consuming the most corrective-maintenance effort per unit. This is a useful pointer for where preventive maintenance scope or spares stocking should be reviewed first.

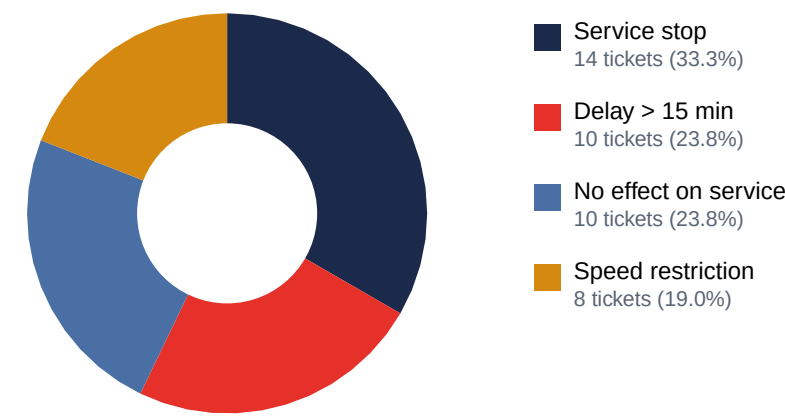
Failure Cause by Individual Train (Top 12 by Ticket Volume)

Train Type - Train Number	NFF	Non-Perm.	Permanent	Total
RG-450 - RG-450-108	1	1	3	5
CS-200 - CS-200-104	1	2	1	4
CS-200 - CS-200-108	0	0	3	3
RG-450 - RG-450-102	1	2	0	3
CS-200 - CS-200-107	0	1	2	3
CS-200 - CS-200-103	1	0	1	2
CS-200 - CS-200-106	1	0	1	2
RG-450 - RG-450-107	1	0	1	2
RG-450 - RG-450-106	0	0	2	2
CS-200 - CS-200-105	1	1	0	2
CS-200 - CS-200-110	0	2	0	2
RG-450 - RG-450-101	0	1	1	2

Analysis: Train RG-450 - RG-450-108 recorded the highest ticket count among individual units in this selection, with 5 closed tickets. The 12 trains listed here together account for 32 tickets, so recurring offenders at the top of this list are the best candidates for a targeted reliability review rather than a fleet-wide one.

3. Effect on Service

Share of tickets by their recorded effect on service, from stopping failures to no measurable operational impact.



Analysis: "Service stop" is the most frequently recorded effect on service, accounting for 33.3% of all tickets in the selected period. The remaining 3 effect categories spread across less service-critical outcomes, which is the healthier pattern for fleet availability. A rising share of service-stopping effects over successive reports would be the clearest early warning of a fleet-wide reliability issue.

Effect on Service by Train Type

Column codes: Service stop / Delay > 15 min / Speed restriction / No effect on service.

Train Type	S	D	S	N	Total
CS-200	9	3	6	3	21
RG-450	5	7	2	7	21

Analysis: Train type CS-200 has the highest share of service-impacting effects relative to its own ticket volume among the train types in this selection. Comparing effect mix per train type — rather than raw counts — surfaces which fleet segment is most exposed to service disruption, independent of how many tickets it simply generates. This is the more useful lens for prioritizing where reliability improvements would have the biggest operational payoff.

Effect on Service — Top 12 Individual Trains

Column codes: Service stop / Delay > 15 min / Speed restriction / No effect on service.

Train Type - Train Number	S	D	S	N	Total
RG-450 - RG-450-108	0	2	0	3	5
CS-200 - CS-200-104	3	0	1	0	4
CS-200 - CS-200-108	1	1	1	0	3
RG-450 - RG-450-102	1	1	0	1	3
CS-200 - CS-200-107	2	0	0	1	3
CS-200 - CS-200-103	1	1	0	0	2
CS-200 - CS-200-106	0	0	2	0	2
RG-450 - RG-450-107	1	1	0	0	2
RG-450 - RG-450-106	0	0	0	2	2
CS-200 - CS-200-105	1	1	0	0	2

Train Type - Train Number	S	D	S	N	Total
CS-200 - CS-200-110	0	0	1	1	2
RG-450 - RG-450-101	0	1	0	1	2

Analysis: Train CS-200 - CS-200-104 recorded the most service-stopping tickets among the individual units listed here, with 3 out of its 4 total tickets. A single train repeatedly producing stopping failures is a stronger maintenance signal than the same count spread across many different units, and is a good candidate for a dedicated root-cause review.

4. Depot Analysis

Distribution of closed tickets by maintenance depot, showing where servicing workload is concentrated.



Analysis: Milano Fiorenza handled the largest share of closed tickets among all depots, at 57.1% of the total. Depot-level volume reflects both fleet allocation and local maintenance workload, so a depot that is disproportionately high relative to the trains assigned to it may need additional resourcing or a review of local procedures.

Depot Breakdown by Failure Cause

Depot	NFF	Non-Perm.	Permanent	Total
Milano Fiorenza	8	6	10	24
Torino Smistamento	3	4	11	18

Analysis: Depot-level failure-cause splits make it possible to see whether one location has a disproportionate share of permanent failures or NFF outcomes compared with the fleet average, which can point to local diagnostic practices or workshop conditions rather than the trains themselves.

5. Top Trains by Failure Count

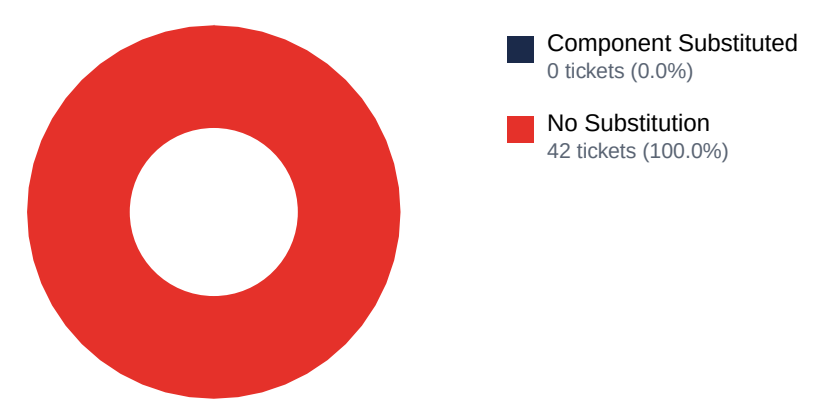
The individual train units with the highest number of closed tickets in the selected period, identified by train type and train number, useful for spotting recurring offenders.

Train Type - Train Number	Total Tickets	Permanent Failures	Substitutions
RG-450 - RG-450-108	5	3	0
CS-200 - CS-200-104	4	1	0
CS-200 - CS-200-108	3	3	0
RG-450 - RG-450-102	3	0	0
CS-200 - CS-200-107	3	2	0
CS-200 - CS-200-103	2	1	0
CS-200 - CS-200-106	2	1	0
RG-450 - RG-450-107	2	1	0
RG-450 - RG-450-106	2	2	0
CS-200 - CS-200-105	2	0	0
CS-200 - CS-200-110	2	0	0
RG-450 - RG-450-101	2	1	0
RG-450 - RG-450-103	2	0	0
RG-450 - RG-450-105	1	0	0
CS-200 - CS-200-109	1	1	0

Analysis: Train RG-450 - RG-450-108 recorded the highest ticket count among individual units in this selection, with 5 closed tickets. The 15 trains listed here together account for 36 tickets, so recurring offenders at the top of this list are the best candidates for a targeted reliability review rather than a fleet-wide one.

6. Component Substitutions

Proportion of tickets that resulted in a component substitution, broken down by train type.



Analysis: Component substitutions were recorded in 0.0% of all closed tickets in this selection, based on part/serial-number references and substitution language found in the corrective action text. This ratio is a useful proxy for spares consumption and can be compared across reporting periods to flag emerging component-reliability trends.

Substitutions by Train Type

Train Type	Substitutions	Total Tickets	% Substituted
CS-200	0	21	0.0%
RG-450	0	21	0.0%

Analysis: Train type CS-200 has the highest component-substitution rate at 0.0% of its tickets, ahead of the other train types in this selection. A consistently high substitution rate for one train type, compared to the others, points to a specific component or subsystem that may be worth a targeted reliability investigation or a design-level corrective action rather than routine repair.